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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not required.
Subject Demographics
Age: not required.
Weight: not required.
Randomization
For SHOxi, we probed with g-P32 dATP randomly primed amplicons generated with custom primers.
Blinding
not detected.
Power Analysis
not detected.
Replication
Counts were averaged and standard deviation calculated between replicates.
Data Information
Availability: RNA-seq data All raw reads and processed data from these experiments are available at the National Center for Biotechnology Information under GEO accession number GSE158891.

Identifiers: Samples were harvested at 0,15,30, and 60 minutes post-actD, extracted for RNA, and malic enzyme mRNA levels were measured with qRT-PCR at 0, 15, 30, and 60 minutes post-actD, in DSHOxi and wild type, under oxidative stress.	0,15,30
Identifiers: RNA-seq data All raw reads and processed data from these experiments are available at the National Center for Biotechnology Information under GEO accession number GSE158891.	GSE158891

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Recombinant DNA			
Five hundred base pairs upstream and downstream of SHOxi, including small overhangs (30 bp), were PCR amplified, the overhangs were annealed together, and then cloned into the integration vector pTA131 to build the knockout plasmid construct.	pTA131		
Overexpression experiments A variant of the overexpression plasmid pTA1228 45 was built to prevent the introduction of an ATG at the beginning of SHOxi.	pTA1228		
Binding site mutagenesis experiments Mutations within the stem-loop binding site of SHOxi were constructed using the Q5 Site-directed Mutagenesis kit (NEB) standard protocol on the previously described SHOxi overexpression construct (in pTA1300).	pTA1300		
Software and Algorithms			
Strand-specific libraries were prepared using the SMART-seq Ultralow RNA input kit (Takara), insert sizes checked with the Bioanalyzer RNA pico kit (Agilent), and either paired-end sequenced (2 x 150 bp) or single-end sequenced (100 bp) on the Illumina HiSeq 2500 platform at the Johns Hopkins University Genetic Resources Core Facility (GRCF).	Agilent		Suggestion: Agilent 4100 MP-AES (RRID:SCR_019390)(link)
The program featureCounts was used to rapidly count reads that map to the NCBI H. volcanii annotation. featureCounts was run with strand-specific options on, paired-end mode on or off, multi-mapping off.	featureCounts		Suggestion: featureCounts (RRID:SCR_012919)(link)
The read counts were then used in the R differential expression software package DESeq2 46.	DESeq2		Suggestion: DESeq, RRID:SCR_000154(link)
Differential expression was calculated using ImageJ.	ImageJ		Suggestion: ImageJ, RRID:SCR_003070(link)
Options used were no-seed, 42degC temperature, no START.	START		Suggestion: START (RRID:SCR_009394)(link)

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is***

not a substitute for expert review. SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq